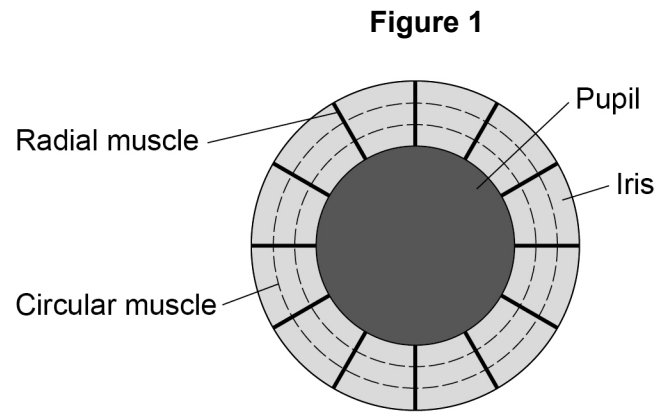


0 1

The iris in the human eye is a muscular structure. The iris changes the size of the pupil.

Figure 1 shows the muscles in the iris.



0 1 . 1

Suggest and explain how the interaction between the muscles labelled in **Figure 1** could cause the pupil to constrict (narrow).

[2 marks]

0	1	2
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The fovea of the eye of an eagle has a high density of cones. An eagle focuses the image of its prey onto the fovea.

Explain how the fovea enables an eagle to see its prey in detail.

Do **not** refer to colour vision in your answer.

[3 marks]

0	1	3
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The retina of the human eye has an area of approximately $1.094 \times 10^3 \text{ mm}^2$

The circular fovea in a human eye has a diameter of $3 \times 10^3 \text{ }\mu\text{m}$

Calculate the area of the fovea as a percentage of the area of the retina.

The area of a circle is πr^2 . Use $\pi = 3.14$ in your calculation.

Show your working.

[2 marks]

Answer _____ %

Question 1 continues on the next page

Turn over ►

[3 marks]

